$u_1^{dc}(t)$

$u_2^{dc}(t)$

$u_{ac}^{m}(t)$

$i_1^{dc}(t)$

$i_2^{dc}(t)$

$i_{ac}^{ph}(t)$

$u_2^{dc}(t) \rightarrow u_2^{dc}\big|_{ref} \rightarrow \varphi_{dab}(t)$

$i_s^{dab}(t) \rightarrow i_s^{dab}\big|_{\min} \rightarrow d_1^{dab}(t) / d_2^{dab}(t)$

$u_1^{dc}(t)$

$u_1^{ac}(t) = d_1^{dab}(t)\, u_1^{dc}(t) \cos(2\pi f_{sw} t)$

$u_2^{ac}(t) = d_2^{dab}(t)\, u_2^{dc}(t) \cos\left[2\pi f_{sw} t + \varphi_{dab}(t)\right]$

$U_1^{ac}(t) = d_1^{dab}(t)\, u_1^{dc}(t)$

$U_2^{ac}(t) = d_2^{dab}(t)\, u_2^{dc}(t)$

$i_{ac}^{ph}(t) \rightarrow i_{ac}^{ph}\big|_{ref}$

$u_{ac}^{m}(t) \rightarrow u_{ac}^{m}\big|_{ref}$

$u_u(t) = U_u \cos\left[\vartheta(t)\right] \rightarrow \breve{\vartheta}_u(t)$

C_2^{dc}

$i_{C_2}^{dc}(t)$

L_{Fu}^{ac}

C_{Fu}^{ac}

$i_{CFu}^{ac}(t)$

$i_s^{ac}(t)$

C_1^{dc}

$i_{C_1}^{dc}(t)$

L_{Fu}^{dc}

C_{Fu}^{dc}

$i_{CFu}^{dc}(t)$

$i_s^{dc}(t)$

$$P_{nom} = \frac{u_1^{dc}\big|_{nom}\, u_2^{dc}\big|_{nom}\, n_1 / n_2}{2\pi f_{sw} L_s} \varphi_{dab} \left(1 - \frac{|\varphi_{dab}|}{\pi}\right)$$

$$f_s = \frac{1}{2\pi\sqrt{L_s C_s}} \rightarrow f_{sw} / f_s \approx 5$$

$u_1^{ac}(t)$

$u_2^{ac}(t)$

L_s

$i_s^{dab}(t)$

$$u_u(t) = U_u \cos\left[\vartheta(t)\right]$$

$$u_v(t) = U_v \cos\left[\vartheta(t) - \frac{2\pi}{3}\right]$$

$$u_w(t) = U_w \cos\left[\vartheta(t) + \frac{2\pi}{3}\right]$$

$$\left.\begin{aligned} u_u(t) &= U_u \cos\left[\vartheta(t)\right] \\ u_u(t) &= U_u \cos\left[\vartheta(t) - \frac{2\pi}{3}\right] \\ u_u(t) &= U_u \cos\left[\vartheta(t) + \frac{2\pi}{3}\right] \end{aligned}\right\} \rightarrow \breve{\vartheta}(t)$$

$u_u(t)$

$u_v(t)$

$u_w(t)$

$i_u(t)$

$i_v(t)$

$i_w(t)$

$\rightarrow \breve{\vartheta}(t)$

$\rightarrow \breve{\vartheta}_v(t)$

$\rightarrow \breve{\vartheta}_w(t)$

$\underline{\underline{u}}_g(t) = \begin{pmatrix} u_u & u_v & u_w \end{pmatrix}^T$

$\underline{\underline{i}}_g(t) = \begin{pmatrix} i_u & i_v & i_w \end{pmatrix}^T$

$\underline{\underline{u}}_{\xi\eta}^{p}(t) = \begin{pmatrix} u_\xi^p & u_\eta^p \end{pmatrix}^T$

$\underline{\underline{u}}_{\xi\eta}^{n}(t) = \begin{pmatrix} u_\xi^n & u_\eta^n \end{pmatrix}^T$

$\underline{\underline{i}}_{\xi\eta}^{p}(t) = \begin{pmatrix} i_\xi^p & i_\eta^p \end{pmatrix}^T$

$\underline{\underline{i}}_{\xi\eta}^{n}(t) = \begin{pmatrix} i_\xi^n & i_\eta^n \end{pmatrix}^T$

$\underline{\underline{u}}_{\xi\eta}^{p}(t)$

$\underline{\underline{u}}_{\xi\eta}^{n}(t)$

$\underline{\underline{i}}_{\xi\eta}^{p}(t)$

$\underline{\underline{i}}_{\xi\eta}^{n}(t)$

$\breve{\vartheta}(t)$

$\left(\breve{\vartheta} \quad \underline{\underline{u}}_{\xi\eta}^{p} \quad \underline{\underline{u}}_{\xi\eta}^{n} \quad \underline{\underline{i}}_{\xi\eta}^{p} \quad \underline{\underline{i}}_{\xi\eta}^{n}\right)$

$\left(\breve{\vartheta} \quad \underline{\underline{u}}_{\xi\eta}^{p}\big|_{avg} \quad \underline{\underline{u}}_{\xi\eta}^{n}\big|_{avg} \quad \underline{\underline{i}}_{\xi\eta}^{p}\big|_{avg} \quad \underline{\underline{i}}_{\xi\eta}^{n}\big|_{avg}\right)$

$\underline{\bar{U}}_1^{ac}$

$\underline{\bar{I}}_s^{dab}$

$\underline{\bar{U}}_2^{ac}$

$j\omega_{sw} L_s \underline{\bar{I}}_s^{dab}$

$\varphi_{dab}(t)$

$u_1^{ac}(t) = U_1^{ac}(t) \cos(\omega_{sw} t)$

$u_2^{ac}(t) = U_2^{ac}(t) \cos\left[\omega_{sw} t + \varphi_{dab}(t)\right]$

$U_1^{ac}(t) = d_1^{dab}(t)\, u_1^{dc}(t)$

$U_2^{ac}(t) = d_2^{dab}(t)\, u_2^{dc}(t)$

$u_g^{dc}(t)$

$\underline{\bar{i}}_{ac}^{ph} = i_{ac}^{ph}\big|_{\mathrm{Re}} + j\, i_{ac}^{ph}\big|_{\mathrm{Im}}$